Week 5 Lab – u3231515

Task 1.1

```python
# task 1.1

import numpy as np
arr = np.arange(10, 20)

print("Array:", arr)
print("Shape:", arr.shape)
print("Dimension:", arr.ndim)
print("Data type:", arr.dtype)
```

Output

```
PS E:\GitHub\week5> python task1_1.py
Array: [10 11 12 13 14 15 16 17 18 19]
Shape: (10,)
Dimension: 1
Data type: int64
```

Task 1.2

```python
# task 1.2

import numpy as np
arr = np.zeros((3, 3), dtype=int)

arr[1, 1] = 7

print(arr)
```

Output

```
PS E:\GitHub\week5> python task1_2.py
[[0 0 0]
 [0 7 0]
 [0 0 0]]
```

Task 1.3

```python
# task 1.3

import numpy as np
arr = np.array([
    [10, 20, 30],
    [40, 50, 60],
    [70, 80, 90]
])

print("Second row:", arr[1])
print("Third column:", arr[:, 2])
print("Submatrix (first 2 rows and cols):", arr
[:2, :2])
```

Output

```
PS E:\GitHub\week5> python task1_3.py
Second row: [40 50 60]
Third column: [30 60 90]
Submatrix (first 2 rows and cols): [[10 20]
 [40 50]]
```

Task 1.4

```python
import numpy as np
arr = np.ones((2, 2, 3), dtype=int)

arr[1, 0, 2] = 5

print(arr)
```

Output

```
[[[1 1 1]
  [1 1 1]]

 [[1 1 5]
  [1 1 1]]]
```

Task 1.5

```python
import numpy as np
arr = np.array([2, 4, 6, 8, 10])

print(arr[::2])
```

Output

```
PS E:\GitHub\week5> python task1_5.py
[ 2  6 10]
```

Task 1.6

```python
# task1_6.py

import numpy as np

a = np.array([5, 10, 15])
b = np.array([2, 4, 6])

print("Sum:", a + b)
print("Difference:", a - b)
print("Product:", a * b)
print("Division:", a / b)
```

Output

```
PS E:\GitHub\week5> python task1_6.py
Sum: [ 7 14 21]
Difference: [3 6 9]
Product: [10 40 90]
Division: [2.5 2.5 2.5]
```

Task 1.7

```python
# task1_7.py

import numpy as np

matrix = np.arange(1, 10).reshape(3, 3)
vector = np.array([1, 0, -1])

result = matrix * vector
print(result)
```

Output

```
PS E:\GitHub\week5> python task1_7.py
[[ 1  0 -3]
 [ 4  0 -6]
 [ 7  0 -9]]
```

Task 1.8

```python
# task1_8.py

import numpy as np

arr = np.arange(0, 16, 3)
print(np.sqrt(arr))
```

Output

```
PS E:\GitHub\week5> python task1_8.py
[0.         1.73205081 2.44948974 3.         3.46410162 3.87298335]
```

Task 1.9

```python
# task1_9.py

import numpy as np

temps = np.array(
    [
        [[10, 15, 20], [25, 30, 35]],
        [[11, 16, 21], [26, 31, 36]],
    ]
)

print(temps[:, 1, :])
```

Output

```
PS E:\GitHub\week5> python task1_9.py
[[25 30 35]
 [26 31 36]]
```

Task 1.10

```python
# task1_10.py

import numpy as np

growth_factors = np.array([0.7, 0.2, 0.5])
rainfall = np.array(
    [
        [40, 50, 60],
        [20, 35, 25],
        [30, 40, 55],
    ]
)

height_increase = growth_factors[:, np.newaxis] *
rainfall
print(height_increase)
```

Output

```
PS E:\GitHub\week5> python task1_10.py
[[28.   35.   42. ]
 [ 4.    7.    5. ]
 [15.   20.   27.5]]
```

Task 2.1

```python
# task2_1.py

import cv2

image = cv2.imread(r"assets/1_Gammar
2021_1_2021_06_03-13-19-23-856.PNG")
print("Type of the image array:", type(image))

if image is not None:
    print("Shape of the image array:", image.shape)
else:
    print("Image could not be read.")
```

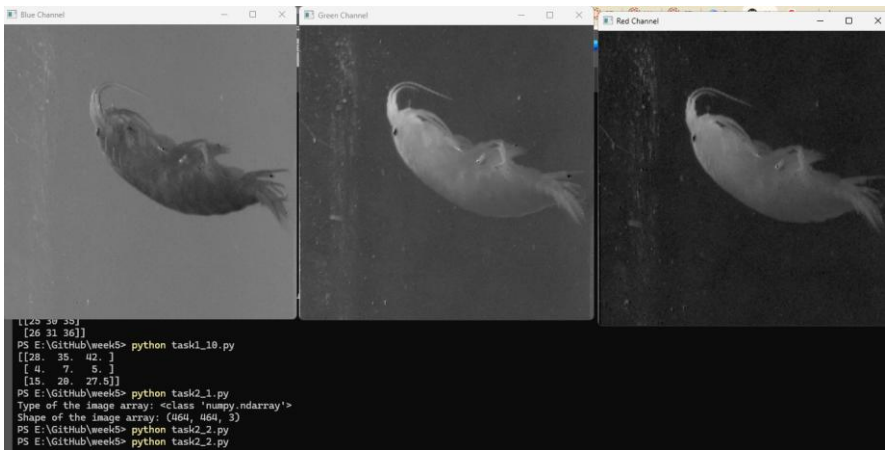
Output

```
PS E:\GitHub\week5> python task2_1.py
Type of the image array: <class 'numpy.ndarray'>
Shape of the image array: (464, 464, 3)
```

Task 2.2

```python
# task2_2.py

import cv2

image = cv2.imread(r"assets/1_Gammar
2021_1_2021_06_03-13-19-23-856.PNG")
b, g, r = cv2.split(image)
cv2.imshow("Blue Channel", b)
cv2.imshow("Green Channel", g)
cv2.imshow("Red Channel", r)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

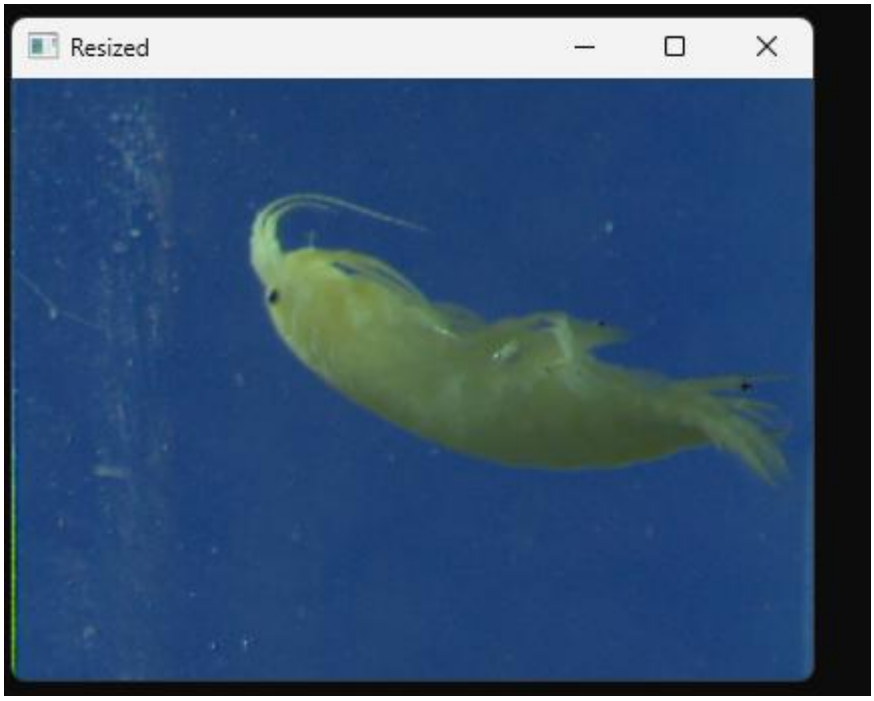
Output



Task 2.3

```
# task2_3.py

import cv2

image = cv2.imread(r"assets/1_Gammar
2021_1_2021_06_03-13-19-23-856.PNG")
resized = cv2.resize(image, (400, 300))
cv2.imshow("Resized", resized)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

Output



Task 2.4

```
# task2_4.py

import cv2

image = cv2.imread(r"assets/1_Gammar
2021_1_2021_06_03-13-19-23-856.PNG")
gray_image = cv2.cvtColor(image, cv2.
COLOR_BGR2GRAY)
cv2.imwrite("gray_image.png", gray_image)
```

Output

Task 2.5

```
# task2_5.py

import cv2

image = cv2.imread(r"assets/1_Gammar
2021_1_2021_06_03-13-19-23-856.PNG")
image_copy = image.copy()
cv2.circle(image_copy, (100, 100), 50, (0, 0,
255), 3)
cv2.imshow("Image with Circle", image_copy)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

Output

Task 2.6

```python
# task2_6.py

import cv2

image = cv2.imread(r"assets/1_Gammar
2021_1_2021_06_03-13-19-23-856.PNG")
print("Original pixel value:", image[50, 50])
image[50, 50] = [255, 255, 255]
cv2.imwrite("modified.png", image)
```

Output

```
PS E:\GitHub\week5> python task2_6.py
Original pixel value: [118  69  34]
```

modified.png

Task 2.7

```python
# task2_7.py

import cv2
import numpy as np

image = cv2.imread(r"assets/1_Gammar
2021_1_2021_06_03-13-19-23-856.PNG")
inverted = 255 - image
combined = np.hstack((image, inverted))
cv2.imshow("Original and Inverted", combined)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

Output

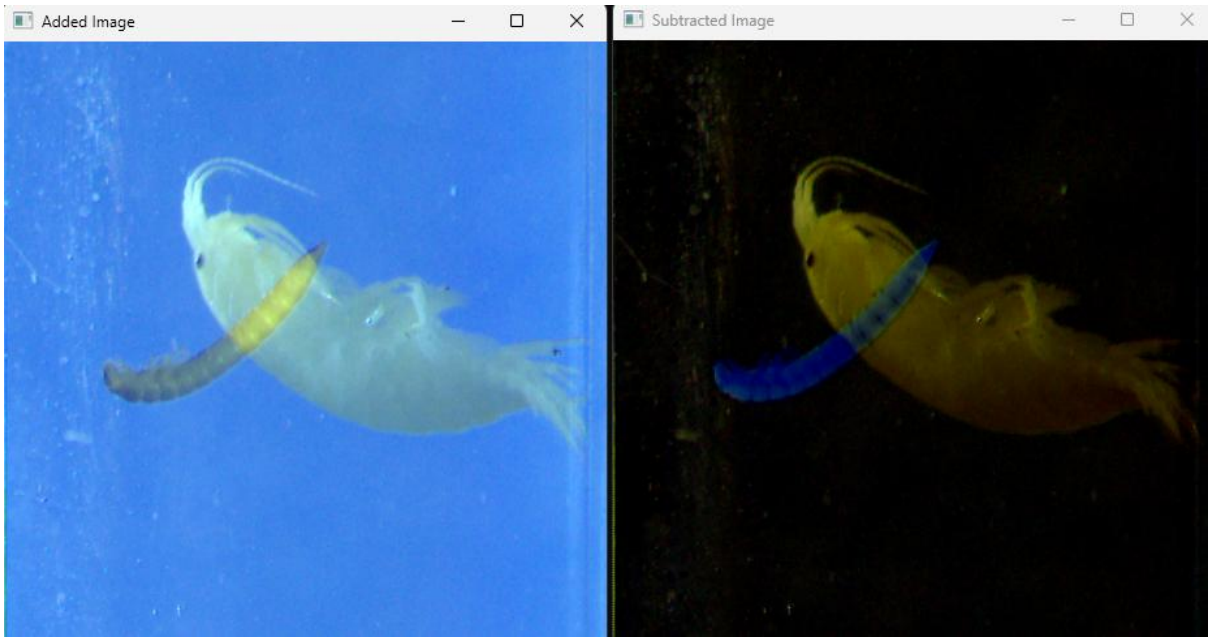
Task 2.8

```
# task2_8.py

import cv2

img1 = cv2.imread(r"assets/1_Gammar
2021_1_2021_06_03-13-19-23-856.PNG")
img2 = cv2.imread(r"assets/1_Limniu
2021_1_2021_06_02-13-08-37-941.PNG")
img2_resized = cv2.resize(img2, (img1.shape[1],
img1.shape[0]))
blended = cv2.addWeighted(img1, 0.6, img2_resized,
0.4, 0)
cv2.imshow("Blended Image", blended)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

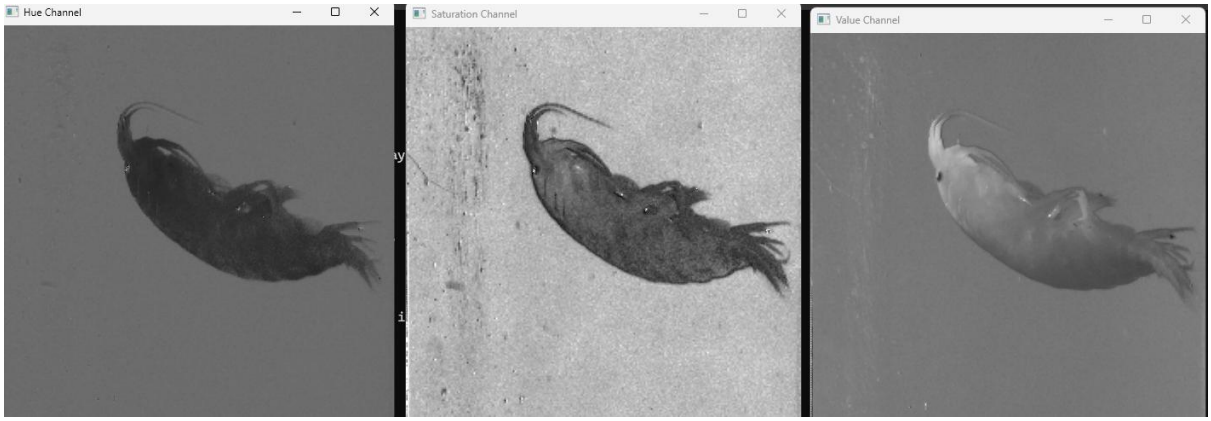
Output

Task 2.9

```python
# task2_9.py

import cv2

img1 = cv2.imread(r"assets/1_Gammar
2021_1_2021_06_03-13-19-23-856.PNG")
img2 = cv2.imread(r"assets/1_Limniu
2021_1_2021_06_02-13-08-37-941.PNG")
img2_resized = cv2.resize(img2, (img1.shape[1],
img1.shape[0]))

added = cv2.add(img1, img2_resized)
subtracted = cv2.subtract(img1, img2_resized)

cv2.imshow("Added Image", added)
cv2.imshow("Subtracted Image", subtracted)
cv2.waitKey(0)
cv2.destroyAllWindows()
```



Task 2.10

```python
# task2_10.py

import cv2

image = cv2.imread(r"assets/1_Gammar
2021_1_2021_06_03-13-19-23-856.PNG")
hsv_img = cv2.cvtColor(image, cv2.COLOR_BGR2HSV)
h, s, v = cv2.split(hsv_img)
cv2.imshow("Hue Channel", h)
cv2.imshow("Saturation Channel", s)
cv2.imshow("Value Channel", v)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

Output